

1 Solution — GIAM 7.3

Problem 5

1.1 Problem

Prove that if 10 points are placed inside a square of side length 3, there will be 2 points within $\sqrt{2}$ of one another.

1.2 Pigeonhole Framing

The problem hands us 10 “things” and asks for a collision under a *geometric* relation (distance $\leq \sqrt{2}$). The standard pigeonhole move here is to partition the 3×3 square into smaller regions of *diameter* at most $\sqrt{2}$, and to have *fewer* regions than points.

Pigeons. The 10 points. **Pigeonholes.** The 9 unit-side cells of a 3×3 grid that tiles the big square.

Since $10 > 9$, Theorem 7.3.1 forces two of the points to lie in the *same* unit cell. The diameter of a unit cell — the maximum distance between two of its points — is $\sqrt{2}$ (its diagonal). Thus those two points are within $\sqrt{2}$ of each other.

The remainder of the writeup makes each step precise and checks tightness.

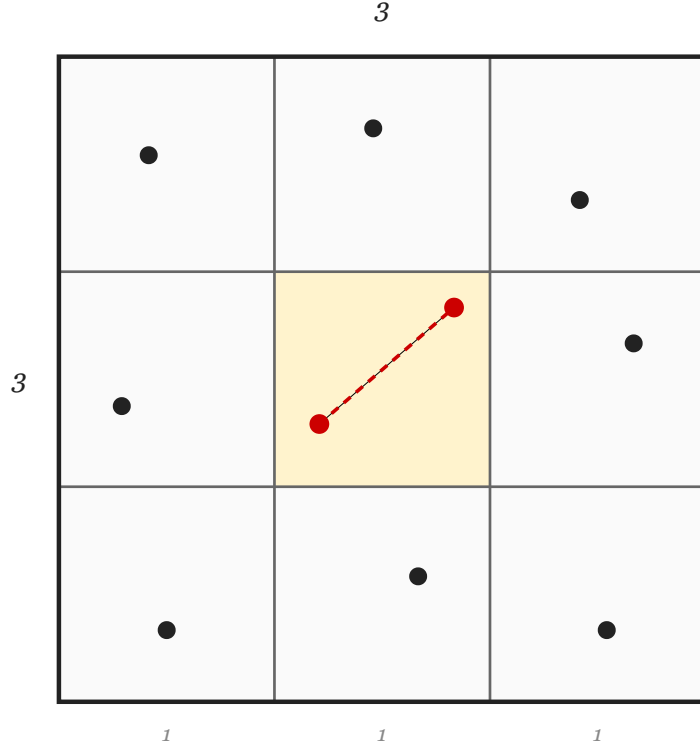


Figure 1.1: The big square has side length 3 and is partitioned into a 3×3 grid of 9 unit cells (each of side 1). With 10 points distributed across the cells, the pigeonhole principle forces some cell — here the shaded center cell — to contain at least two points. The dashed segment connects two such points; the argument shows their distance is at most $\sqrt{2}$.

1.3 Step 1 — Partition the Big Square

Place the 3×3 square in the plane as

$$Q = [0, 3] \times [0, 3].$$

For $i, j \in \{0, 1, 2\}$, define the unit cells

$$R_{ij} = [i, i + 1) \times [j, j + 1)$$

with the convention that the right- and top-boundary cells $R_{2,j}, R_{i,2}$ include the boundary $x = 3$ resp. $y = 3$ (so that every point of Q lies in exactly one R_{ij}). The 9 cells partition Q :

$$Q = \bigsqcup_{i,j \in \{0,1,2\}} R_{ij}, \quad |\{(i,j)\}| = 9.$$

(The half-open convention is a technical nicety to make every point lie in exactly one cell; the geometry — in particular, each cell's diameter — depends only on the *closure* $\overline{R_{ij}} = [i, i + 1] \times [j, j + 1]$, which is a closed unit square.)

1.4 Step 2 — Apply the Pigeonhole Principle

Let $P = \{p_1, p_2, \dots, p_{10}\} \subseteq Q$ be the 10 given points, and define the *cell-assignment* function

$$\phi : P \longrightarrow \{R_{ij}\}_{i,j \in \{0,1,2\}}, \quad \phi(p) = \text{the unique } R_{ij} \text{ containing } p.$$

Then

$$|\text{Dom}(\phi)| = 10 > 9 = |\text{Rng}(\phi)|,$$

so by Theorem 7.3.1, ϕ is not injective: there exist distinct $p, q \in P$ with $\phi(p) = \phi(q)$. That is, *two of the points lie in the same unit cell* R_{ij} .

1.5 Step 3 — Diameter of a Unit Cell is $\sqrt{2}$

It remains to bound the distance between two points lying in the same unit cell. The cell's closure is the closed unit square $\overline{R_{ij}} = [i, i + 1] \times [j, j + 1]$.

For any two points $p = (x_1, y_1)$ and $q = (x_2, y_2)$ in this closed unit square,

$$|x_1 - x_2| \leq 1, \quad |y_1 - y_2| \leq 1,$$

so by the Pythagorean theorem,

$$\|p - q\| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \leq \sqrt{1^2 + 1^2} = \sqrt{2}.$$

The bound is *attained* exactly when p and q are at opposite corners — i.e., $\{(i, j), (i + 1, j + 1)\}$ or $\{(i, j + 1), (i + 1, j)\}$.

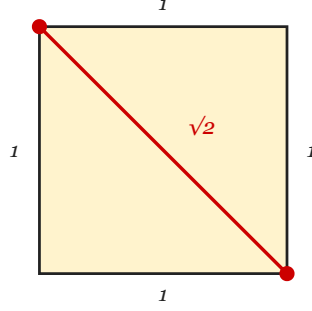


Figure 1.2: Inside any unit cell, the maximum distance between two points is the length of the diagonal, $\sqrt{2} \approx 1.414$, attained when the two points sit at opposite corners.

1.6 Putting It Together

From Steps 2 and 3: the pigeonhole principle gives two of the 10 points in the same unit cell R_{ij} , and any two points in a unit cell satisfy

$$\|p - q\| \leq \sqrt{2}.$$

So two of the 10 points are within $\sqrt{2}$ of one another. ■

1.7 Sharpness — 9 Points Need Not Suffice

The bound is *tight*: there exists a placement of 9 points in Q with *every* pairwise distance strictly greater than $\sqrt{2}$.

Take the 3×3 lattice

$$L = \{0, 1.5, 3\} \times \{0, 1.5, 3\}, \quad |L| = 9.$$

For any two distinct points $p, q \in L$, the coordinate differences are multiples of 1.5; in particular, at least one coordinate differs by ≥ 1.5 . Hence

$$\|p - q\| \geq 1.5 > \sqrt{2} \approx 1.414.$$

So L is a 9-element configuration in Q with no two points within $\sqrt{2}$ — confirming the threshold of 10 in the original statement cannot be improved.

(*Minor point:* L uses points on the boundary of Q . If “inside” is read strictly as “open interior,” perturb each lattice point inward by ε for any $\varepsilon \in (0, (1.5 - \sqrt{2})/2)$ — every pairwise distance remains $> \sqrt{2}$.)

1.8 Remark — The General Pattern

The argument is a special case of the *partition lemma* for pigeonhole-style geometric claims:

If a region Q can be partitioned into n subregions each of diameter $\leq d$, and one places $n + 1$ points in Q , then two of the points are within d of one another.

For an $n \times n$ square partitioned into n^2 unit cells, the lemma gives: $n^2 + 1$ points in $[0, n]^2$ force two within $\sqrt{2}$.

Square side	# unit cells	Forcing threshold	Distance bound
2	4	5	$\sqrt{2}$
3	9	10	$\sqrt{2}$
4	16	17	$\sqrt{2}$
n	n^2	$n^2 + 1$	$\sqrt{2}$

Table 1.1

The distance bound is *always* $\sqrt{2}$ (the unit-cell diagonal); only the threshold scales with the area.

1.9 Remark — Why $\sqrt{2}$? And What Would It Take to Improve It?

The bound $\sqrt{2}$ is *exactly* the diameter of the cells we used to partition. To force a *smaller* distance, one needs a finer partition — and that requires *more* points.

For example, if we partition the 3×3 square into **36** cells of side $\frac{1}{2}$ (each of diameter $\frac{\sqrt{2}}{2} \approx 0.707$), pigeonhole would require **37** points (not **10**) to force two within $\frac{\sqrt{2}}{2}$. The trade-off is universal: finer cells $\rightarrow$ tighter distance bound $\rightarrow$ more points needed.

The choice "9 cells, 10 points, $\sqrt{2}$ " hits the *minimum* point count for which a unit-cell partition gives any forcing. Any smaller partition gives no useful forcing; any larger gives a worse-than- $\sqrt{2}$ bound.

1.10 Remark — The Pigeonhole “Move” in Geometric Problems

This problem (and Problem 6 in the same exercise set — the equilateral-triangle version) follow a recipe:

1. **Geometric goal.** “There exist two points within d of each other.”
2. **Choose a partition.** Of the ambient region into pieces of diameter $\leq d$.
3. **Apply pigeonhole.** $n + 1$ points in n pieces force a co-piece pair.
4. **Read off the distance bound.** Co-piece points are within d .

The creative step is *step 2* — *which* partition? For squares, a grid of smaller squares. For triangles, a grid of smaller triangles (Problem 6). For disks, a partition into pie slices or annular wedges. The diameter of each piece must equal the desired distance bound; the count of pieces must equal one less than the count of points.

The remaining steps are mechanical — the principle handles them once the partition is named.